

Einwirkung

Wind

```
from engicalc import *  
import numpy as np
```

Parameter

```
alpha_r = 0.19  
z_g = 380*m  
z = 6*m  
c_h = 1.6 * ((z/z_g)**alpha_r + 0.375)**2  
Cell(show_expression=False, rows = 4)
```

$$\alpha_r = 0.19 \quad z_g = 380 \text{ m} \quad z = 6 \text{ m} \quad c_h = 1.1$$

Staudruck

```
q_p0 = 0.90*kN/m**2  
q_p = c_h * q_p0  
  
Cell(show_expression=False, rows = 4)
```

$$q_{p0} = \frac{0.9 \text{ kN}}{\text{m}^2} \quad q_p = \frac{0.99 \text{ kN}}{\text{m}^2}$$

Windkraft

```
c_f1 = 1.55  
c_red = 1.00  
c_d = 1.00  
q_k1 = c_red * c_d * c_f1 * q_p  
  
Cell(show_expression=False, rows = 4)
```

$$c_{f1} = 1.55 \quad c_{red} = 1.0 \quad c_d = 1.0 \quad q_{k1} = \frac{1.54 \text{ kN}}{\text{m}^2}$$